The Dataset I Used

Real Estate Sales from 2001-2023

DATA.GOV

Original data was 131 megabytes and had over 1 million rows and 14 columns. I filtered the data set until it had only 6 columns and 78000 rows. The smaller dataset was only 5 megabytes and contained data only from the years 2020 and 2021.

List Year	Assessed Value	Sale Amount	Sales Ratio	Residential Type	Location
2020	133000.00	248400.00	0.5354	Single Family	NT (-73.06822 41.35014)
2021	179990.00	362500.00	0.4965	Condo	79115982 41.773452988)
2020	528490.00	775000.00	0.6819	Single Family	NT (-72.89675 41.79445)
2021	70850.00	130000.00	0.545	Condo	21583028 41.214441988)
2020	286860.00	540000.00	0.5312	Single Family	NT (-72.97635 41.79258)
2021	78890.00	197000.00	0.4004	Condo	24709009 41.654481008)
2021	264260.00	42500.00	6.2178	Single Family	-72.923670995 41.48174)
2021	144200.00	325000.00	0.4436	Single Family	35396005 41.794073014)
2020	83020.00	175000.00	0.4744	Single Family	NT (-72.33294 41.51729)
2020	89580.00	165000.00	0.5429	Condo	NT (-72.84836 41.32761)
2021	294500.00	500000.00	0.589	Condo	572950037 41.71884799)
2020	737590.00	1364000.00	0.5407	Single Family	NT (-73.59662 41.05717)
2021	112140.00	200000.00	0.5607	Single Family	01396003 41.331707999)
2021	76790.00	180000.00	0.4266	Condo	22237019 41.359736999)
2020	166040.00	330000.00	0.5031	Single Family	NT (-72.00456 41.38926)
2021	156530.00	350000.00	0.4472	Single Family	60699023 41.748629018)
2021	201980.00	395000.00	0.5113	Single Family	43289041 41.519865003)
2021	96110.00	238000.00	0.4038	Single Family	07392037 41.691687019)
2021	73570.00	105300.00	0.6986	Condo	99623985 41.269475008)
2021	884650.00	1540000.00	0.5744	Condo	INT (-73.40765 41.11366)
2021	295300.00	495000.00	0.5965	Condo	376168013 41.28292901)
2021	198520.00	326000.00	0.6089	Single Family	62987699 41.579752989)

Numerical Variables

Sale Amount vs. Assessed Value

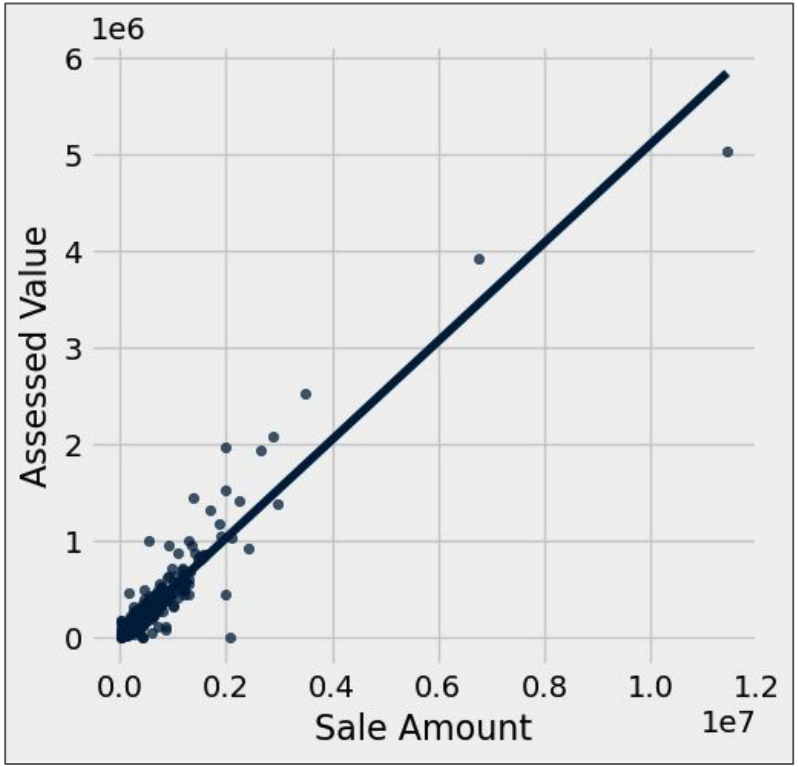
This graph shows the correlation between how much a property was valued at and how much it actually sold for.

R value for this graph: 0.9597

The R value measures the correlation between two numerical variables.

95% CI: [0.8812, 0.9676]

This confidence interval shows the range that 95% of the resampled data fell into. In other words, if you repeatedly recorded this data, the r value between theses two variables would be in that interval 95% of the time.



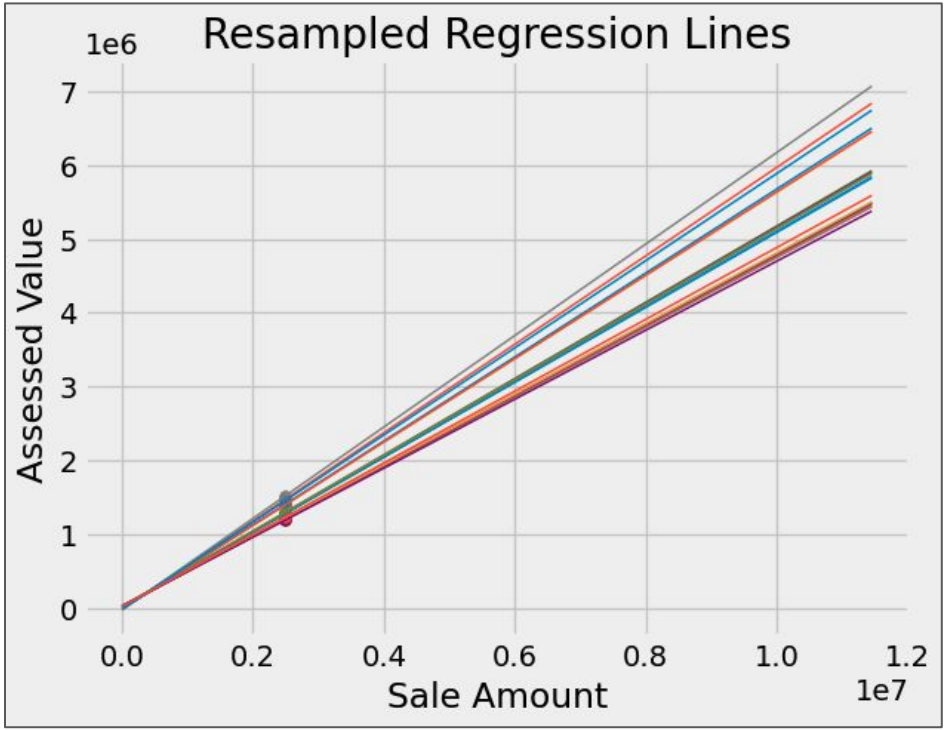
Prediction

Predicting the y value of an arbitrary x value using the dataset.

The arbitrary x value is 2,500,000.

99% CI: [1154932, 1545918]

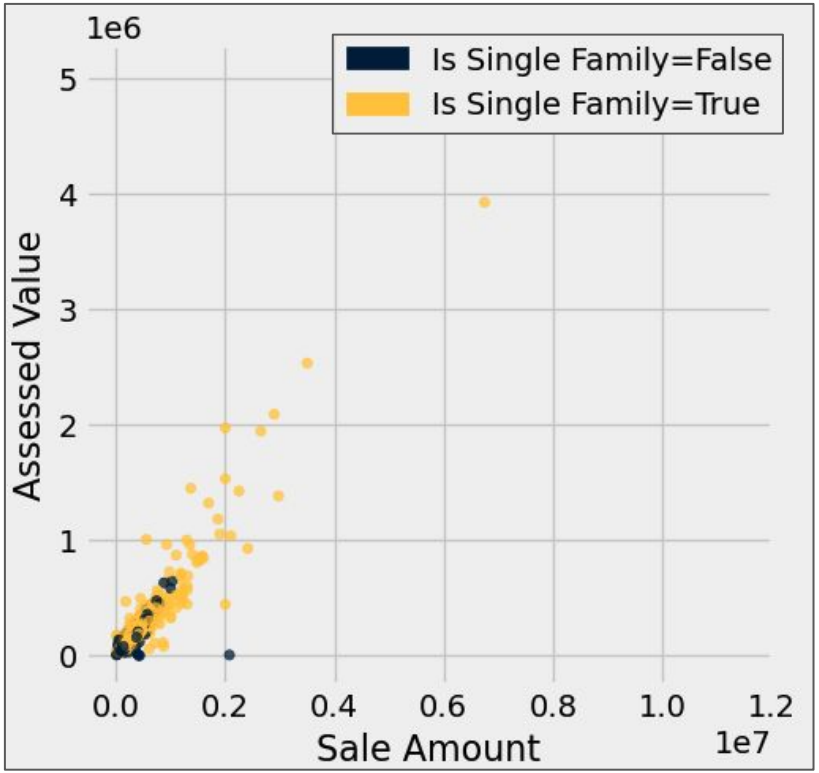
This 99% confidence interval means that 99% of the time, the prediction of the y value will fall between those values.



My Classifier

A classifier is an algorithm that classifies data points into groups.

My classifier will classify a datapoint into two groups, if a house is for a single family or for multiple families. It will do this based on the sale amount and the assessed value.



This graph is the original data, color coded with this distinction.

My Classifier

How I built my classifier:

Step 1: Split the dataset into training and test sets. The training set is used to give the algorithm an example of how it should classify the data and the test set is used to see how accurate the algorithm is.

Step 2: Make the classification algorithm. I used K-Nearest Neighbors which classifies a datapoint based on how the nearest other data points are classified.

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Training set:    750 examples
Test set:        250 examples
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List Year	Assessed Value	Sale Amount	Sales Ratio	Residential Type	Location	Is Single Family
2021	202720	485000	0.4179	Three Family	POINT (-72.923853971 41.330926009)	False
2020	143080	275000	0.520291	Four Family	POINT (-72.71674 41.72965)	False
2021	234700	527000	0.4453	Single Family	POINT (-73.350732985 41.381825996)	True
2021	113190	239900	0.4718	Single Family	POINT (-73.052739982 41.948739999)	True
2021	90020	240000	0.375	Single Family	POINT (-73.074650025 41.578377016)	True

... (745 rows omitted)

List Year	Assessed Value	Sale Amount	Sales Ratio	Residential Type	Location	Is Single Family
2020	125050	190500	0.6564	Single Family	POINT (-72.15268 41.43717)	True
2021	183230	520000	0.3523	Condo	POINT (-72.674149029 41.284836983)	False
2021	134820	315000	0.428	Single Family	POINT (-72.644427974 41.52923199)	True
2021	56460	122000	0.4627	Condo	POINT (-73.008816965 41.468395005)	False
2021	145100	327900	0.4425	Single Family	POINT (-72.931506984 41.911189995)	True

Evaluating my Classifier

To classify the test set data, the classifier compares each point to the points in the training data whose assessed value and sale amount are the closest to that point. If most of those points are single family homes, it will classify it as a single family home and vice versa.

Then we compare each data point in the newly classified test set data to what those data points were classified as originally and see how many the classifier got correct.

Accuracy: 0.66

An accuracy of 0.66 means the classifier got 66% of the classifications correct.

Improving my Classifier

To improve the accuracy of 66%, I compared the ratio between the sale amount and assessed value as well as the sale amount and assessed value in the distance calculation.

New Accuracy: 0.70

Conclusion

Some things I learned about my dataset:

- My dataset does not have much correlation between its different columns other than between sale amount and assessed value.
- My dataset was way too big to use all at once so I ended up taking a random sample at the beginning and using that instead of the entire dataset.

A success rate of 70% is not very good. I do not think you can accurately predict if a property is for a single family or not based on just the sale amount and assessed value.